

Table 1. Description of Variables Used in the Statistical Learning Models

Variable	Type	Description
Student_Type	Categorical	Student category (College, School, Working Student)
Sleep_Hours	Continuous	Average daily sleep duration (hours)
Study_Hours	Continuous	Average daily study duration (hours)
Social_Media_Hours	Continuous	Average daily social media usage (hours)
Attendance	Continuous	Class attendance (%)
Exam_Pressure	Continuous	Examination pressure score
Family_Support	Continuous	Family support score
Month	Categorical	Month of data collection
Stress_Level	Binary	Response variable (Low Stress = 0, High Stress = 1)

Table 2. Predictive Performance of the Statistical Learning Models for Binary Classification

Model	Accuracy	Sensitivity	Specificity	F1	ROC AUC	Brier Score
Logistic Regression	0.8171	0.6081	0.9066	0.6660	0.8634	0.1307
Ridge Regression	0.8134	0.5593	0.9223	0.6426	0.8635	0.1329
LASSO Regression	0.8172	0.6081	0.9068	0.6662	0.8634	0.1307
Elastic Net	0.8171	0.6068	0.9072	0.6656	0.8634	0.1307
Random Forest	0.8162	0.5759	0.9191	0.6527	0.8603	0.1328